

Offline Coefficient-Space Comparison for Case-2 Maps ANN vs GPR, $n_s \in \{131, 141, 151\}$

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May 22, 2026

Objective

This report compares six previously trained Case-2 maps in pure offline reconstruction mode:

$$(\mu, t) \mapsto q_s(\mu, t),$$

without running online nonlinear ROM solves.

The compared models include ANN and GPR maps with

$$n_s \in \{131, 141, 151\},$$

using the same linear PROM reference trajectories.

Reference trajectories are the linear PROM coefficients $q_N^{lin}(t; \mu)$ with $n_{tot} = 151$ at

$$\mu^{(v)} = (4.875, 0.0225), \quad \mu^{(1)} = (4.560, 0.0190), \quad \mu^{(2)} = (5.190, 0.0260).$$

Error definitions

For a model with split $n_{tot} = n_p + n_s$, we compare predicted secondary coefficients against the corresponding tail of the linear reference:

$$q_s^{ref}(t; \mu) = [q_N^{lin}(t; \mu)]_{n_p+1:n_{tot}}, \quad q_s^{pred}(t; \mu) = \mathcal{M}_\theta(\mu, t).$$

For each predicted coefficient index $i = 1, \dots, n_s$:

$$E_i^{abs}(\mu) = \|q_{s,i}^{ref} - q_{s,i}^{pred}\|_2, \quad E_i^{rel}(\mu) = 100 \frac{\|q_{s,i}^{ref} - q_{s,i}^{pred}\|_2}{\|q_{s,i}^{ref}\|_2 + \varepsilon}.$$

Global secondary-trajectory error:

$$E_F(\mu) = 100 \frac{\|Q_s^{ref} - Q_s^{pred}\|_F}{\|Q_s^{ref}\|_F + \varepsilon}.$$

Since models have different n_s , each curve starts at global index $n_p + 1$ with $n_p = n_{tot} - n_s$. Therefore, missing lower global indices for some curves are expected.

Pointwise summary

Table 1: Offline coefficient errors per point (%).

Model	n_s	E_F	$\text{mean}(E_i^{rel})$	$\text{p95}(E_i^{rel})$	$\text{max}(E_i^{rel})$
$\mu = (4.875, 0.0225)$					
ANN	131	11.020	27.578	64.427	77.394
ANN	141	4.968	17.719	40.772	51.943
ANN	151	1.791	30.688	79.297	87.480
GPR	131	1.29e-04	2.95e-04	9.14e-04	1.19e-03
GPR	141	7.87e-05	2.92e-04	9.77e-04	1.27e-03
GPR	151	1.86e-05	2.87e-04	1.02e-03	1.36e-03
$\mu = (4.560, 0.0190)$					
ANN	131	102.964	142.204	201.576	272.048
ANN	141	9.206	27.858	59.705	78.524
ANN	151	3.842	49.200	101.068	118.742
GPR	131	100.142	100.027	100.171	100.573
GPR	141	100.170	100.039	100.265	100.573
GPR	151	100.656	100.093	100.531	103.486
$\mu = (5.190, 0.0260)$					
ANN	131	69.570	113.381	170.202	201.128
ANN	141	10.268	33.535	64.266	80.333
ANN	151	5.474	53.547	105.780	138.958
GPR	131	99.934	99.998	100.051	100.314
GPR	141	99.897	99.989	100.048	100.314
GPR	151	99.744	99.976	100.055	100.539

Average over the three points

Table 2: Averaged offline coefficient errors over the three points (%).

Model	n_s	$\text{mean } E_F$	$\text{mean}(\text{mean}(E_i^{rel}))$	$\text{mean}(\text{p95}(E_i^{rel}))$	$\text{mean}(\text{max}(E_i^{rel}))$
ANN	131	61.185	94.387	145.402	183.523
ANN	141	8.147	26.371	54.914	70.267
ANN	151	3.702	44.479	95.382	115.060
GPR	131	66.692	66.675	66.741	66.963
GPR	141	66.689	66.676	66.772	66.963
GPR	151	66.800	66.690	66.862	68.009

Per-coefficient curves

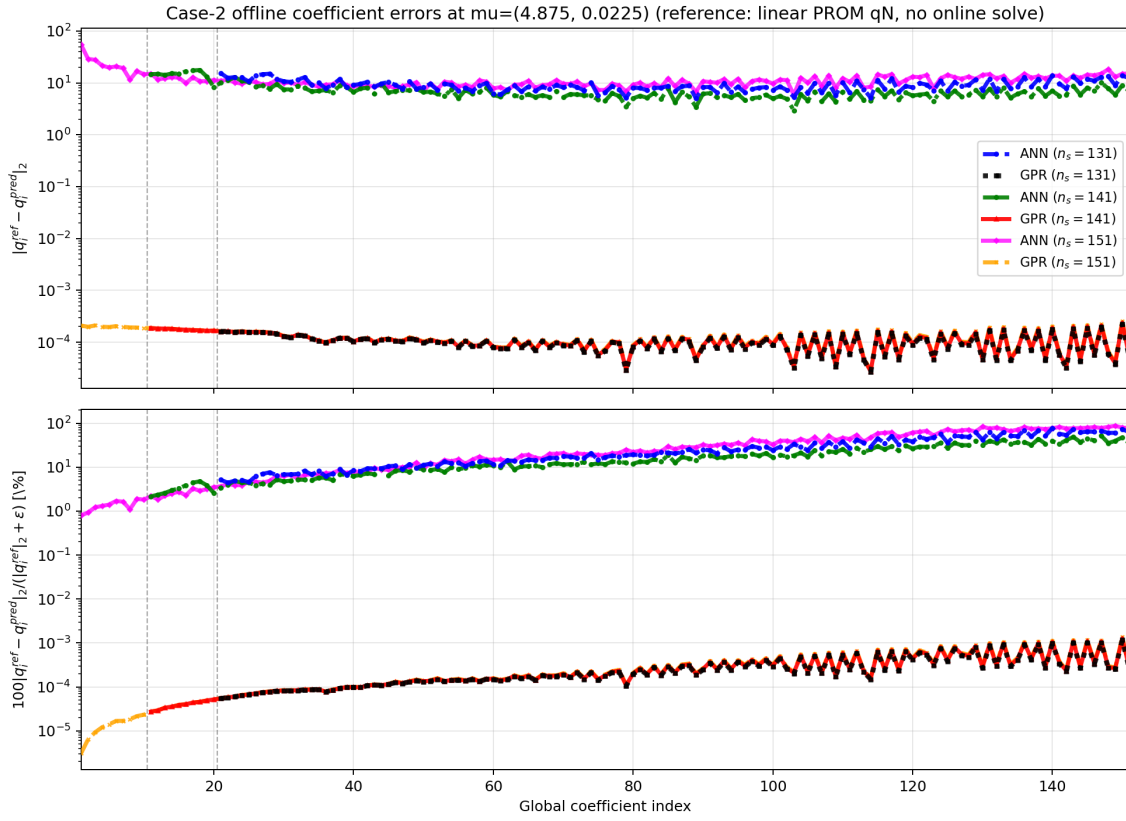


Figure 1: Absolute and relative coefficient errors vs global coefficient index at $\mu^{(v)} = (4.875, 0.0225)$.

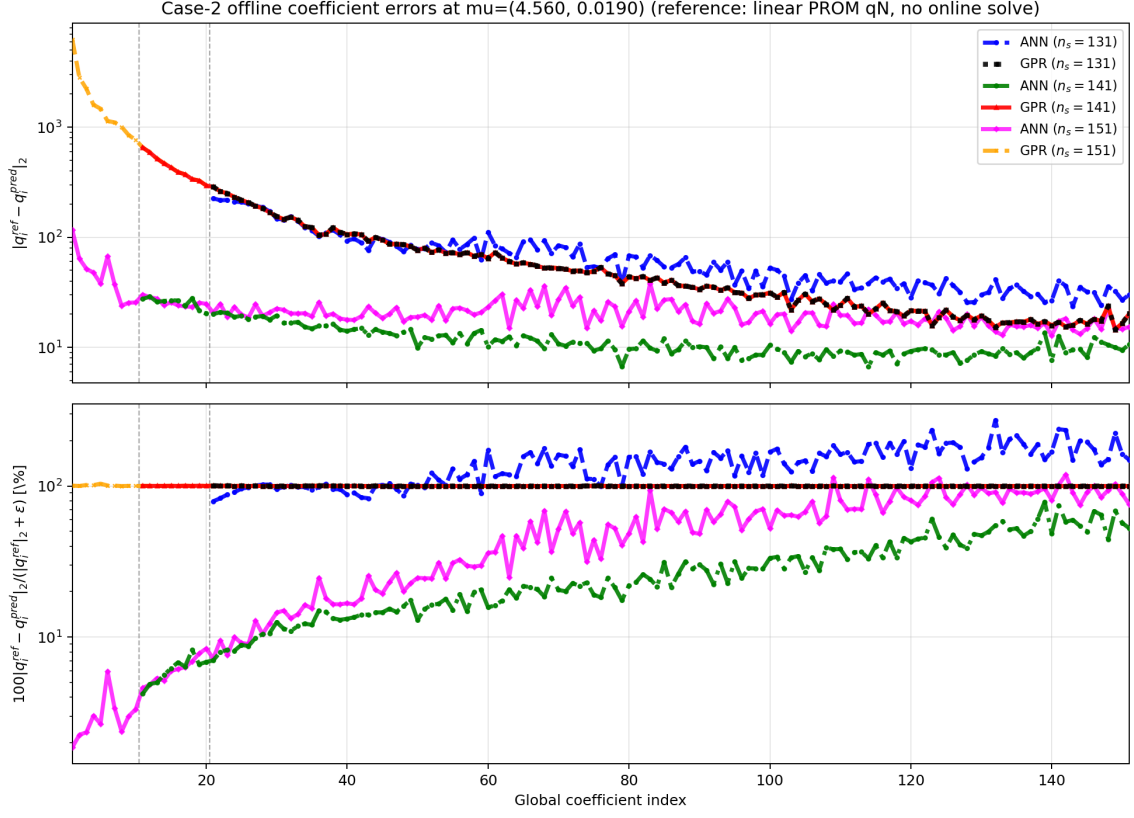


Figure 2: Absolute and relative coefficient errors vs global coefficient index at $\mu^{(1)} = (4.560, 0.0190)$.

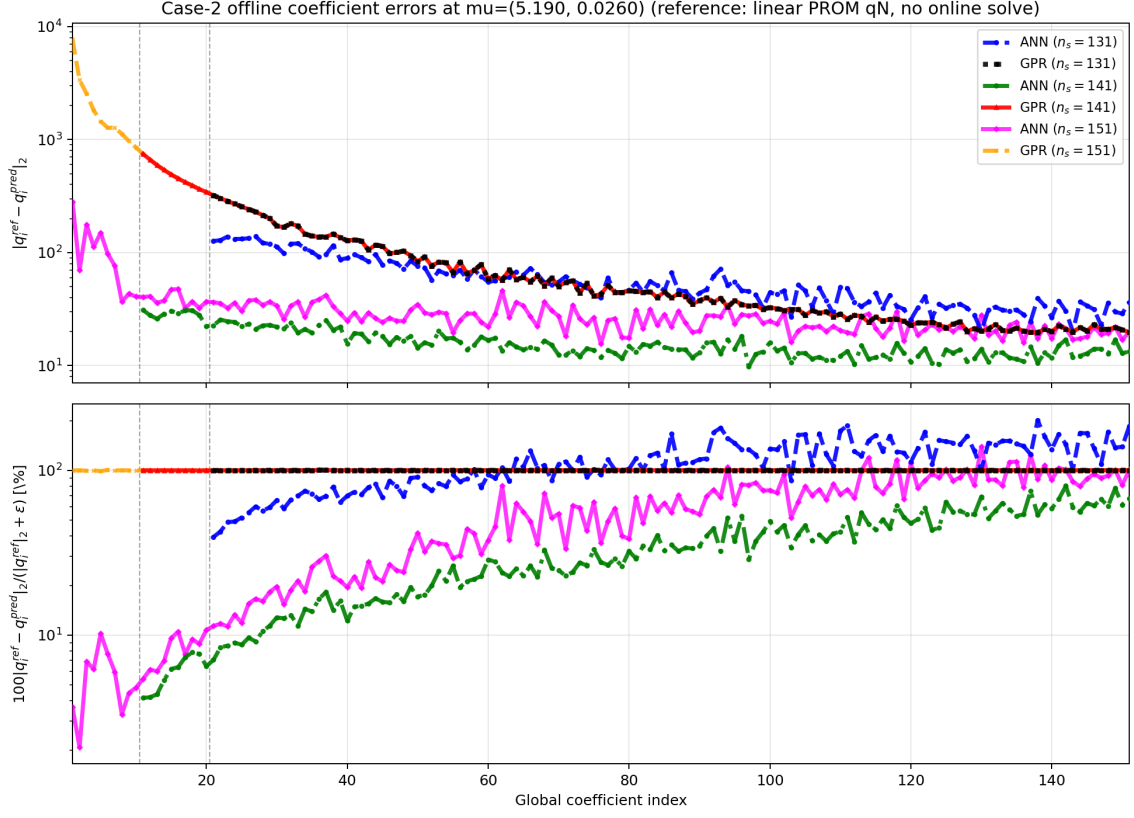


Figure 3: Absolute and relative coefficient errors vs global coefficient index at $\mu^{(2)} = (5.190, 0.0260)$.

GPR zoom

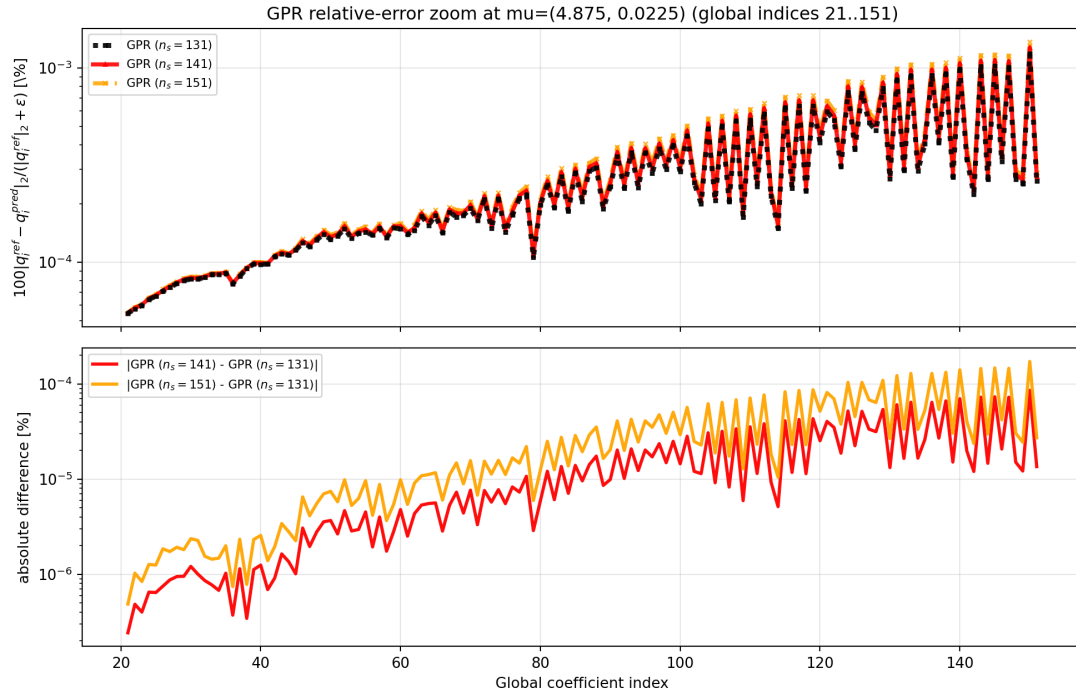


Figure 4: Zoomed GPR comparison at $\mu^{(v)} = (4.875, 0.0225)$, including pairwise differences.

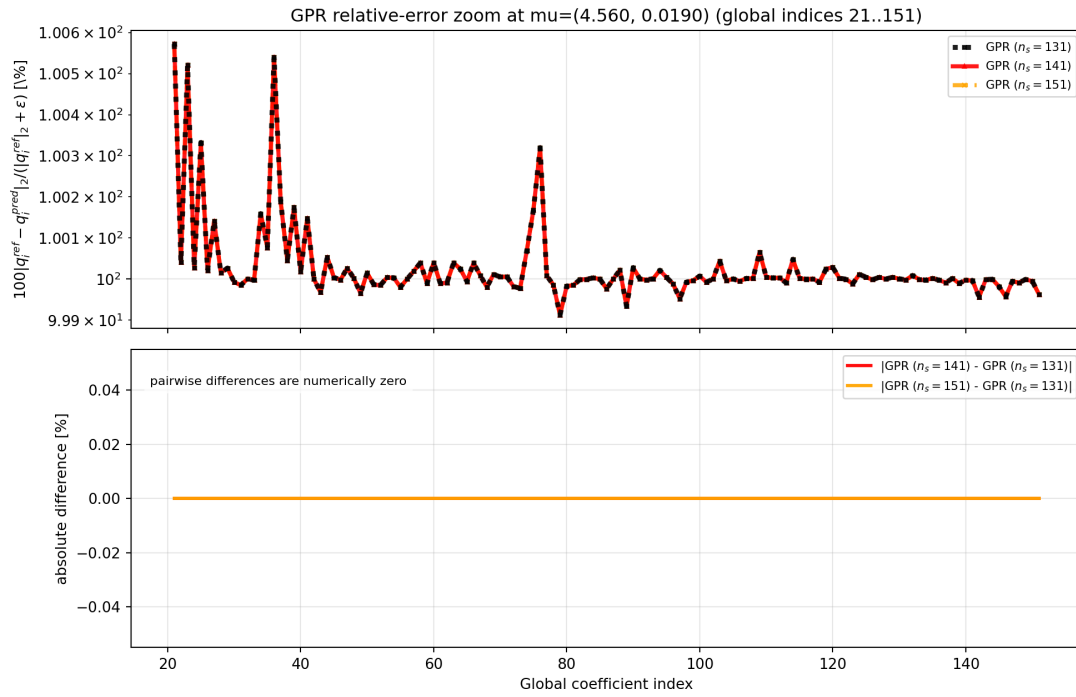


Figure 5: Zoomed GPR comparison at $\mu^{(1)} = (4.560, 0.0190)$, including pairwise differences.

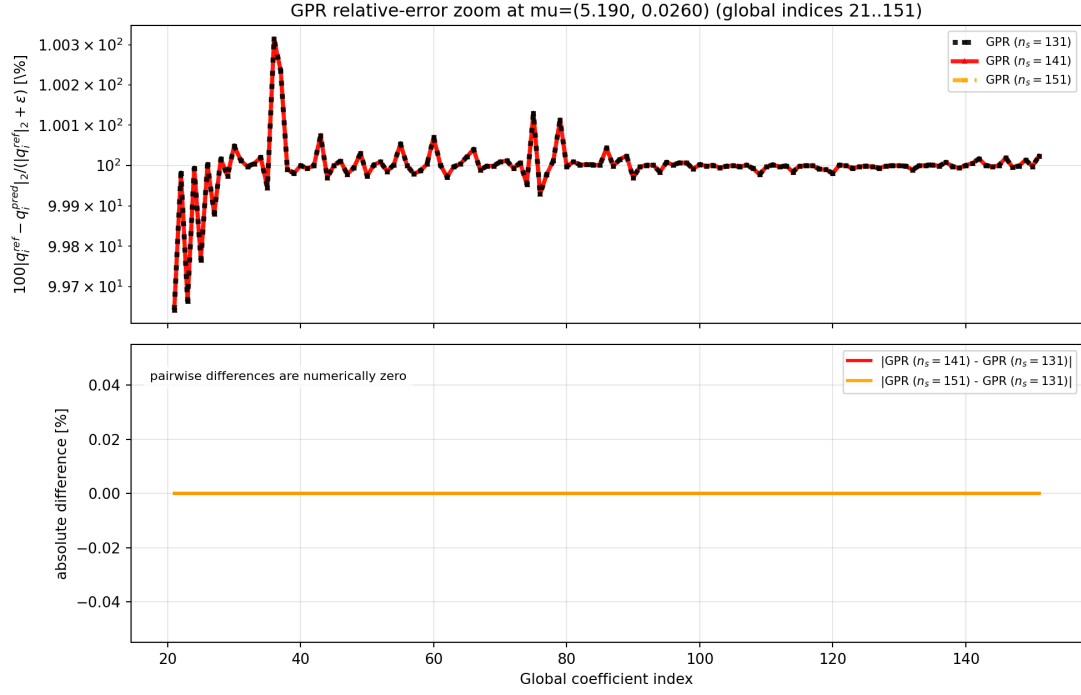


Figure 6: Zoomed GPR comparison at $\mu^{(2)} = (5.190, 0.0260)$, including pairwise differences.

Remarks

This report is strictly an offline map-comparison diagnostic in coefficient space. It does not include online ROM reconstruction of state trajectories.